

Data Flow Diagram

Date	12 July 2026
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

External Entity

User / Data Analyst

Process 1 – Data Collection

- Collect electricity consumption dataset.
 - Verify data accuracy.
 - Prepare data for analysis.
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Process 2 – Data Storage & Processing

- Store dataset in Microsoft SQL Server.
 - Perform SQL queries.
 - Clean and transform data.
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Data Store

Microsoft SQL Server Database

Contains:

- State Details
 - Region Details
 - Electricity Consumption Records
 - Year-wise Data
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Process 3 – Data Visualization

- Generate Tableau Dashboards.
- Create Tableau Story.
- Display interactive visualizations.

Process 4 – Web Application

- Flask retrieves dashboard information.
 - Display dashboards and reports.
 - Provide user-friendly interface.
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Output

- State-wise Consumption Analysis
- Region-wise Analysis
- Monthly Consumption Trends
- Quarterly Analysis
- Year-wise Consumption
- Lockdown Impact Analysis
- Interactive Dashboard
- Tableau Story

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

